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 TOPICAL REVIEW

A Systematic Review of Extended Reality (XR) Technologies in Agriculture and Related Sectors (2022–2024)

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ABSTRACT The agricultural sector confronts critical challenges such as climate change, overpopulation, and resource shortages, necessitating urgent innovative solutions. In this context, Extended Reality (XR) technologies have emerged as a transformative technology, offering groundbreaking approaches to modern farming practices. This study conducts a thorough literature review to explore the integration of XR technologies in agriculture, livestock farming, and aquaculture. By categorizing studies based on focus areas, application types, and human-machine interaction dimensions, it identifies emerging trends, highlights critical gaps, and proposes future direction for advancing these fields. The analysis revealed a significant increase in XR technology adoption over the past two years, highlighting its potential within the sector. The findings suggest that XR technologies can enhance productivity, improve training methods, and facilitate remote collaboration in agricultural settings. However, future research should focus on addressing health, privacy, and security concerns to enhance seamless and immersive XR experiences, thereby maximizing the potential benefits of XR technology in the primary sector. Therefore, these technologies will transform the agriculture industry, fostering a resilient and sustainable food system to meet the demands of a growing global population.

INDEX TERMS Extended reality (XR), augmented reality (AR), virtual reality (VR), mixed reality (MR), human-machine interaction (HCI), immersive experiences, agriculture, livestock farming, aquaculture, education, training.

I. INTRODUCTION

Modern challenges such as climate change [1], overpopulation [2], freshwater scarcity [3], energy poverty [4], urbanization [5], and an aging population [6] pose significant threats to our well-being. The COVID-19 pandemic has further intensified these issues by straining social structures and hindering long-term development and prosperity [7]. Despite these challenges, the reliance on plants and animals remains fundamental to human existence, providing essential resources for dietary needs, clothing, shelter, and health.

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Therefore, agricultural and aquatic products continue to be vital sources of food, fibers, fuels, pharmaceuticals, and other raw materials, significantly contributing to the advancement of human civilization [8].

Extended Reality (XR), which includes Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), offers a promising solution to address the limitations exacerbated by the COVID-19 pandemic and other contemporary challenges. XR technologies aim to enhance human capabilities and experiences through immersive computer-generated environments, with VR creating fully immersive virtual worlds, AR overlaying virtual elements onto the real world, and MR facilitating real-time interaction between physical

and virtual spaces [9], [10], [11], [12]. These technologies encompass a variety of tools, including head-mounted displays (HMDs), portable devices such as smartphones, microphones, gesture recognition systems, position trackers, motion controllers, data gloves, and monitors, all designed to enhance human-machine interaction across VR, AR, and MR environments [13]. They have demonstrated successful applications in diverse fields including medicine [14], manufacturing [15], education [16], and training [17]. Specifically, in medicine, XR supports education, diagnosis, and procedural planning. In manufacturing, they facilitate tasks such as assembly, quality control, and design evaluation [18]. They also foster innovative learning experiences in education and improve human-robot collaboration and safety protocols in training [16].

The ongoing advancement and increasing accessibility of XR technology are set to drive its wider adoption in agriculture, leading to smarter, more efficient, and sustainable farming practices. XR technologies are already transforming agriculture by enhancing precision farming, reducing resource overuse, and minimizing environmental impacts [19]. The potential applications of XR in agriculture are extensive, providing real-time data and training opportunities that boost efficiency and productivity, particularly in remote areas [20]. Moreover, XR contributes to sustainability by facilitating precision agriculture and reducing resource consumption [17].

The integration of XR into agriculture represents a groundbreaking advancement with the potential to revolutionize farming practices. By providing real-time, precise information and enabling remote training and assistance, XR technology can significantly enhance productivity, reduce costs, and promote sustainability within the agricultural sector [21], [22], [23]. This advancement is crucial as it not only addresses immediate operational challenges but also aligns with broader goals of improving global food security and environmental conservation.

This study explores the transformative potential of XR in agriculture, livestock farming and aquaculture, highlighting its ability to address modern challenges and contribute to global food security and environmental conservation. Despite its promising applications, XR in these fields is still emerging, with notable trends and gaps that warrant further investigation. The study emphasizes the need for additional research to address health, privacy, and security concerns while improving interaction and immersion in XR environments, ultimately facilitating greater adoption of XR technology and achieving substantial advancements in the primary sector.

To structure this investigation, the study aims to:

- 1) Identify key applications and investigations of XR in agriculture, livestock farming, and aquaculture.
- 2) Analyze trends in XR research within these domains in recent years.
- 3) Examine the role of human-computer interaction in agricultural solutions based on XR.

- 4) Evaluate the benefits, limitations, and challenges of XR adoption in agriculture.
- 5) Identify research gaps and future directions of XR in the agricultural sector.

To achieve the objectives, the research systematically employs a robust methodological framework, including systematic search query, meticulous results filtering, and thorough article classification, followed by detailed statistical analysis. The results section offers a cumulative analysis of research study quantities, investigating XR applications across various dimensions, and evaluating human-computer interaction, including XR technology types and viewer types. In the discussion, the study explores XR research in agriculture, assesses the benefits and challenges of XR adoption, and examines its environmental and economic impacts. The analysis also addresses ethical and privacy concerns, identifies future research gaps, and highlights opportunities for cross-disciplinary collaboration and innovation.

II. MATERIALS AND METHODS

A. RESEARCH QUESTIONS

To systematically investigate the role of XR technologies in agriculture, this study addresses the following key research questions:

- 1) What are the primary applications and studies of XR technologies in agriculture, livestock farming, and aquaculture?
- 2) How has XR research evolved in these fields over time?
- 3) How does human-computer interaction influence the effectiveness of XR-based agricultural applications?
- 4) What are the main challenges and limitations hindering the adoption of XR in agriculture?
- 5) What are the key research gaps and future opportunities for XR in the agricultural sector?

By addressing these questions, the study aims to provide a comprehensive understanding of the current state of XR applications in agriculture, highlighting potential opportunities and challenges of these technologies.

B. SEARCH QUERY

A comprehensive search strategy was implemented utilizing multiple academic search engines, including ScienceDirect (<https://www.sciencedirect.com>), IEEE Xplore (<https://www.ieee.org>), Google Scholar (scholar.google.com), ResearchGate (<https://www.researchgate.net>), and Scopus (<https://www.scopus.com>), to identify relevant research articles for the analysis. The specific queries used in each search engine are detailed in Table 1, ensuring a thorough and systematic selection of studies for inclusion in this research.

C. RESULTS FILTERING

The scope of this study was defined by applying specific criteria to the literature search. Publications were restricted to the period from January 2022 to December 2024, ensuring a focus on the most recent research. The study

TABLE 1. Search engines and queries were employed for the scope of this study.

Search Engine	Website	Query
ScienceDirect	https://www.sciencedirect.com	TITLE-ABS-KEY(("mixed reality" OR "virtual reality" OR "augmented reality" OR "extended reality" OR "AR" OR "VR" OR "XR" OR "MR") AND ("agriculture"))
IEEE Xplore	https://www.ieee.org	((("Document Title":"mixed reality" OR "Document Title":"virtual reality" OR "Document Title":"augmented reality" OR "Document Title":"extended reality" OR "Abstract":"mixed reality" OR "Abstract":"virtual reality" OR "Abstract":"augmented reality" OR "Abstract":"extended reality") AND ("Document Title":"agriculture" OR "Document Title":"farming" OR "Document Title":"livestock" OR "Document Title":"crops" OR "Document Title":"smart farming" OR "Document Title":"smart agriculture" OR "Document Title":"precision agriculture" OR "Document Title":"precision farming" OR "Document Title":"poultry" OR "Document Title":"fishery" OR "Document Title":"aquaculture" OR "Document Title":"plant" OR "Document Title":"farm" OR "Document Title":"leaf" OR "Document Title":"cultivation" OR "Document Title":"agrotourism" OR "Abstract":"agriculture" OR "Abstract":"farming" OR "Abstract":"livestock" OR "Abstract":"crops" OR "Abstract":"smart farming" OR "Abstract":"smart agriculture" OR "Abstract":"precision agriculture" OR "Abstract":"precision farming" OR "Abstract":"poultry" OR "Abstract":"fishery" OR "Abstract":"aquaculture" OR "Abstract":"plant" OR "Abstract":"farm" OR "Abstract":"leaf" OR "Abstract":"cultivation" OR "Abstract":"agrotourism"))
ResearchGate	https://www.researchgate.net	("mixed reality" OR "virtual reality" OR "augmented reality" OR "extended reality" OR "AR" OR "VR" OR "XR" OR "MR") AND ("agriculture" OR "farming" OR "livestock" OR "crops" OR "smart farming" OR "smart agriculture" OR "precision agriculture" OR "precision farming" OR "poultry" OR "fishery" OR "aquaculture" OR "plant" OR "farm" OR "leaf" OR "cultivation" OR "agrotourism")
Scopus	https://www.scopus.com	TITLE-ABS-KEY(("mixed reality" OR "virtual reality" OR "augmented reality" OR "extended reality" OR "AR VR" OR "XR" OR "MR") AND ("agriculture" OR "farming" OR "livestock" OR "crops" OR "smart farming" OR "smart agriculture" OR "precision agriculture" OR "precision farming" OR "poultry" OR "fishery" OR "aquaculture" OR "plant" OR "farm" OR "leaf" OR "cultivation" OR "agrotourism")) AND (LIMIT-TO(SUBJAREA, "AGRI") OR LIMIT-TO(SUBJAREA, "ENVI") OR LIMIT-TO(SUBJAREA, "ENGI") OR LIMIT-TO(SUBJAREA, "COMP"))

focuses on articles, conference papers and book chapters, as these document types typically represent primary research findings. The subject areas were confined to agriculture, environment, engineering, and computer science, aligning with the interdisciplinary nature of the topic. To maintain consistency and accessibility, English-language publications were mostly considered for this study. These parameters were established to capture the most relevant and up-to-date research in the field.

The initial search yielded a substantial corpus of 981 articles. Following the PRISMA framework for systematic reviews [23], we implemented a rigorous filtering process to refine our selection. The first round of screening involved evaluating titles and abstracts to exclude articles that were irrelevant to our research objectives (e.g., studies on virtual fish) and duplicate entries. This initial filtering process eliminated 810 articles, leaving us with 171 potentially relevant papers. Subsequently, we conducted a more thorough examination of the full manuscripts. This second round of screening excluded papers that were inaccessible in their entirety or fell outside the scope of our study. This process further reduced our pool by 110 articles. Ultimately, our systematic review incorporated 61 research studies, comprising both journal articles and conference papers, which formed the foundation for our analysis (Figure 1).

D. ARTICLE CLASSIFICATION

In this research, we have organized the articles into two distinct tables: 1) development-focused articles, and

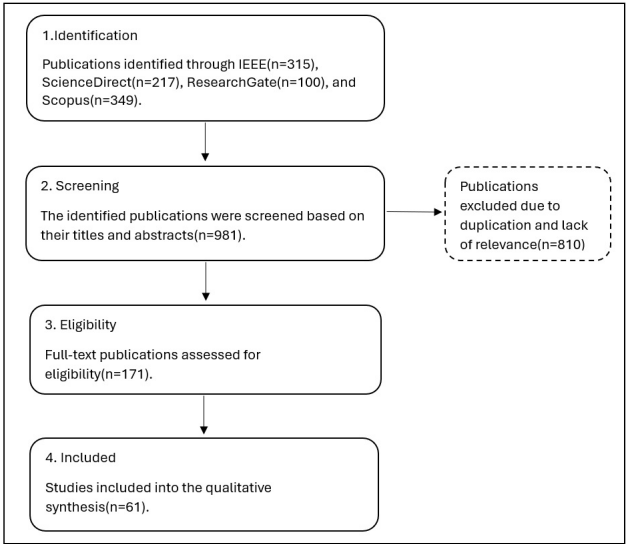


FIGURE 1. The PRISMA flowchart outlining the literature review search process for this study.

2) non-development-focused research studies. The development-focused articles were categorized into two main topics, further divided into five sub-topics relevant to the aims of this review. The first two sub-topics were organized based on the use of XR technology, specifically in terms of focus area type and application type. This categorization provides insight into how XR technology is applied across different domains and practical implementations. The remaining

TABLE 2. Classification of development-focused Articles across various topics covered in this research.

Use		Human Machine Interaction		
Focus Area	Application	XR Type	XR Viewer Device	XR Controller
Agriculture	Decision	AR	Portable Device	Handheld
Livestock Farming	Action	MR	Monitor	Handheld Motion
Aquaculture	System Development	VR	Head Mounted Display	Hand Tracking
Auxiliary/Education	Education			Wearable
				Voice
				Head Mounted
				Portable Device
				Personal Computer
				Controllers Equipment

TABLE 3. Classification of non-development-focused Research Studies various topics covered in this research.

Use		Human Machine Interaction
Paper Type	Focus Area	XR Type
Book Chapter	Agriculture	AR
Research Paper	Livestock Farming	MR
Review Article	Aquaculture	VR
Opinion Article	Auxiliary/Education	
Case Study		

three sub-topics concentrated on aspects of human-machine interaction. The third sub-topic categorized research studies according to the type of XR technology used. The fourth and fifth sub-topics classified studies based on the XR viewer devices and XR control devices, respectively. This classification scheme highlights the hardware and interface elements essential for XR experiences.

For a detailed overview of the classification types used in this study, please refer to Table 2. This table offers a structured breakdown of the various categories employed in the analysis of the development-focused articles, facilitating a comprehensive understanding of both the technological applications and human-centric aspects of XR research. This classification was derived from related reviews [11], [22], [24], [25], [26], [27], [28]. On the other hand, non-development-focused research studies were categorized into main topics, which were further divided into three subtopics aligned with the objectives of this review, including categories like paper type, focus area, and XR technology. This hierarchical classification scheme allowed for a systematic organization of the literature, facilitating a comprehensive analysis of the field. By structuring the information in this manner, we were able to identify trends and interrelationships across various aspects of XR research. Table 3 provides an overview of the classification types used in this study.

E. STATISTICAL ANALYSIS

In our statistical analysis, we first quantified the number of research studies published within each classification type. We then conducted a detailed examination of the distribution across focus areas, applications, XR types, viewing devices, and input devices. To gain deeper insights, we investigated the relative contributions of various XR types, viewing devices, and input devices within each focus area and application type. Lastly, we employed simple tabulated correspondence

analysis to elucidate the relationships between focus areas and application types in conjunction with XR types, viewing devices, and input devices. This multi-faceted approach allowed for a comprehensive understanding of the trends and patterns within the reviewed literature.

III. RESULTS

As outlined in Table 4, this study examines 40 development-focused articles to assess the impact of XR technology on production-related sectors, particularly focusing on decision-making processes and actions within agriculture, livestock management, and aquaculture. In addition, 21 publications of non-development-focused research studies, such as review articles, book chapters, and conference papers, are analyzed to provide a comprehensive understanding of the field (Table 5). The study further classifies these research efforts based on focus area, application, type of XR, XR viewing device, and XR controller. Detailed discussions of these classifications are illustrated through tables and charts to offer a thorough overview.

A. CUMULATIVE ANALYSIS OF RESEARCH STUDY QUANTITIES

According to Tables 4 and 5, more than half of the research studies were conducted between 2023 and 2024. Specifically, 48 studies (approximately 78.69%) were published during this period. Out of these, 15 studies (about 24.59%) occurred in 2024 alone, compared to the 13 studies (approximately 21.31%) from 2022 to 2023. Most research studies were published in journals, with 29 papers (about 47.45%) appearing in research journals, whereas 28 (approximately 45.9%) were published as conference papers. Other types of publications, including book chapters, technical reports, and theses, accounted for 4 studies (about 6.65%). Figure 2 and

TABLE 4. Classification of Development-focused Articles.

Publication	Year	Focus Category	Application	Publication Type	XR Type	Viewing Device
[29]	2024	Auxiliary/Education	Education	Journal	VR	Monitor
[30]	2024	Auxiliary/Education	Education	Journal	VR	HMD, Monitor
[31]	2024	Agriculture	Decision	Journal	AR	Portable Device, Monitor
[32]	2024	Agriculture	Education	Conference	VR, AR	HMD, Monitor
[33]	2024	Aquaculture	Education	Journal	VR	HMD
[34]	2024	Agriculture	Education	Conference	VR	HMD
[35]	2024	Auxiliary/Education	Education	Book Chapter	AR, VR	Portable Device
[36]	2023	Auxiliary/Education	Education	Journal	AR	Portable Device
[37]	2023	Agriculture	Monitoring Action	Conference	AR	Portable Device
[38]	2023	Agriculture	Decision	Conference	VR	HMD
[39]	2023	Agriculture	System Development	Journal	VR	HMD, Monitor
[40]	2023	Auxiliary/Education	Education	Conference	VR	HMD, Monitor
[41]	2023	Agriculture	Decision	Journal	AR	Portable Device
[42]	2023	Agriculture	System Development	Conference	AR	Portable Device
[43]	2023	Agriculture	System Development	Journal	AR	Portable Device, Monitor
[44]	2023	Agriculture	Decision	Conference	VR	HMD, Monitor
[45]	2023	Auxiliary/Education	Education	Conference	AR	Portable Device
[46]	2023	Livestock Farming	System Development	Journal	AR	HMD
[47]	2023	Agriculture	System Development	Journal	AR	Portable Device
[48]	2023	Agriculture	Decision	Journal	VR	HMD
[49]	2023	Agriculture	Education	Conference	VR	HMD
[50]	2023	Agriculture	Decision	Conference	AR	Portable Device
[51]	2023	Agriculture	Monitoring Action	Conference	AR	HMD
[52]	2023	Auxiliary/Education	Education	Conference	MR	HMD
[53]	2023	Agriculture	Decision	Conference	AR	HMD
[54]	2023	Agriculture	Monitoring Action	Conference	AR	HMD
[55]	2023	Agriculture	Monitoring Action	Journal	AR	Portable Device
[56]	2023	Livestock Farming	Decision	Journal	AR	HMD
[57]	2023	Aquaculture	Monitoring Action	Journal	AR	Portable Device
[58]	2023	Auxiliary/Education	Education	Journal	VR	HMD
[59]	2023	Auxiliary/Education	Education	Journal	AR	Portable Device
[60]	2023	Agriculture	Monitoring Action	Conference	AR	HMD
[61]	2023	Auxiliary/Education	Education	Thesis	VR	HMD
[62]	2022	Agriculture	Decision	Conference	AR	Portable Device
[63]	2022	Agriculture	System Development	Conference	AR	Portable Device
[64]	2022	Agriculture	System Development	Conference	AR	Portable Device
[65]	2022	Agriculture	Monitoring Action	Conference	AR	Portable Device
[66]	2022	Agriculture	Monitoring Action	Conference	AR	Portable Device
[67]	2022	Agriculture	Monitoring Action	Conference	AR	HMD
[68]	2022	Agriculture, Aquaculture	System Development	Journal	AR	Portable Device

Figure 3 show the distribution of published research studies by type and the percentage distribution of these study types.

B. ANALYZING XR APPLICATIONS BY FOCUS AREA

In reviewing research studies on XR technology applications in production-related fields, both distinct differences and similarities become apparent. The findings reveal that most of the studies, approximately 63.4%, focused on agriculture. Livestock farming and aquaculture accounted for about 4.9% and 7.3%, respectively. Notably, 24.4% of the studies were related to educational applications, including agricultural education and workforce development. The analysis of research distribution by application area highlights a growing interest in precision agriculture and smart farming over the past three years, compared to the previous decade. Both agriculture and auxiliary focus areas have garnered significant attention regarding XR technology applications

since the three years of the millennium, compared with the other focus areas. Figure 4 illustrates these findings.

In contrast to our findings, [22] analyses studies from 2001 to 2021, highlighted a lower focus on agriculture at 41.3%, while livestock farming and aquaculture had slightly higher representation at 12.7% and 7.9%, respectively. Furthermore, 38.3% of the research concentrated on auxiliary topics (Figure 5).

C. ANALYZING XR USAGE BY APPLICATION TYPE

In the context of XR technology applications, most studies focused on educational purposes, accounting for approximately 35.0% of the total. The next largest areas of the application were decision-making and monitoring actions, each comprising about 22.5%. System development represented around 20.0%. These findings suggest that XR technology is predominantly used in educational settings, with notable

TABLE 5. Classification of non-development-focused research studies.

Publication	Year	Paper Type	Focus Category	Publication	XR Technology
[69]	2024	Review Article	Agriculture	Conference	AR
[70]	2024	Research Paper	Agriculture	Journal	XR
[71]	2024	Review Article	Agriculture	Conference	AR
[72]	2024	Review Article	Livestock farming	Journal	AR
[73]	2024	Research Paper	Livestock Farming	Journal	XR
[74]	2024	Research Paper	Auxiliary/Education	Journal	VR
[75]	2024	Review Article	Auxiliary/Education	Journal	VR
[22]	2023	Review Article	Agriculture, Livestock Farming, Aquaculture	Journal	XR (AR, MR, VR)
[76]	2023	Review Article	Agriculture	Journal	Not applicable
[77]	2023	Review Article	Agriculture	Conference	XR
[78]	2023	Opinion Article	Agriculture	Journal	XR
[79]	2023	Book Chapter	Agriculture	Book Chapter	XR
[80]	2023	Review Article	Agriculture	Journal	XR
[81]	2023	Research Paper	Agriculture	Journal	VR
[82]	2023	Opinion Article	Auxiliary/Education	Technical Report	VR
[83]	2022	Case Study	Auxiliary/Education	Conference	VR
[84]	2022	Review Article	Agriculture	Conference	AR
[85]	2022	Review Article	Agriculture	Conference	AR
[86]	2022	Research Paper	Agriculture	Conference	AR
[87]	2022	Research Paper	Auxiliary/Education	Journal	VR
[88]	2022	Research Paper	Livestock Farming	Journal	AR

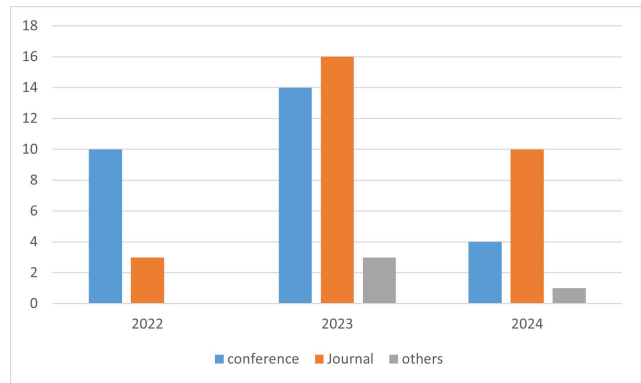


FIGURE 2. Distribution of published research studies by type.

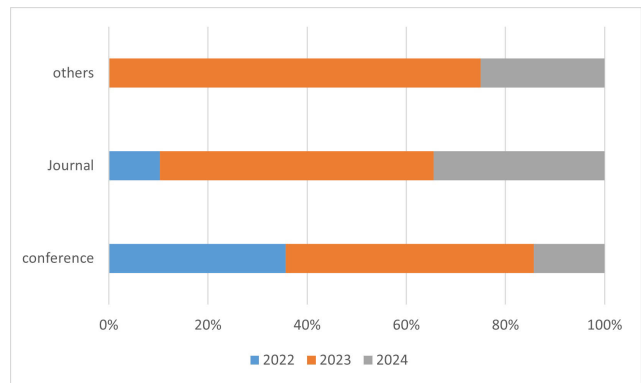


FIGURE 3. Distribution percentage of published research study types.

applications in decision-making, system development, and monitoring (Figure 6).

D. HUMAN-COMPUTER INTERACTION WITHIN AGRICULTURAL XR RESEARCH

1) XR TECHNOLOGY TYPES

A comparative analysis of application development trends in immersive technologies indicates that AR is the most

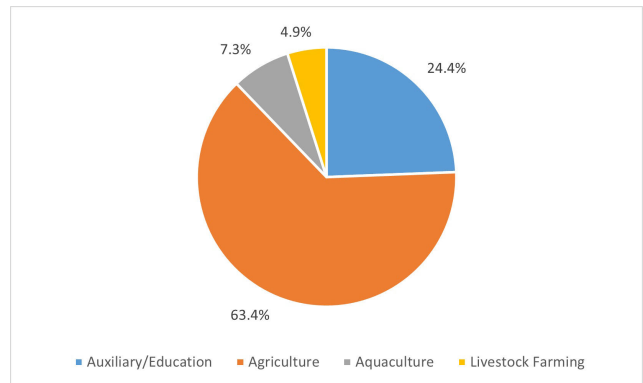


FIGURE 4. XR technology use per focus area (2022-2024).

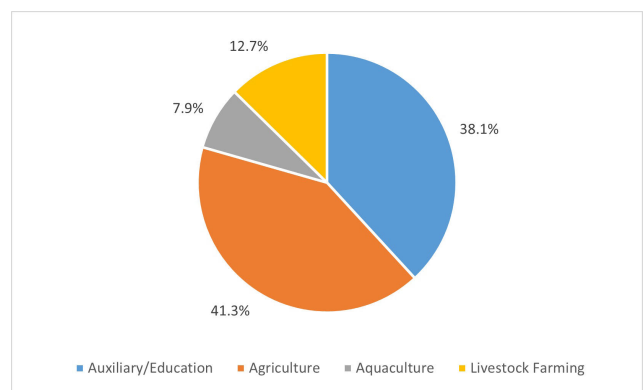


FIGURE 5. XR technology use per focus area (2001-2021) [22].

prevalent type utilized across the various focus areas of this study, comprising approximately 63.4% of the total. Following AR, VR accounts for about 34.1%. MR is represented in only 2.4% of the studies. Conversely, [22] examines publications from 2001 to 2021, and reveals a different distribution: VR leading with 54.5%, AR at 41.8%, and MR showing a slight increase to 3.6%. This

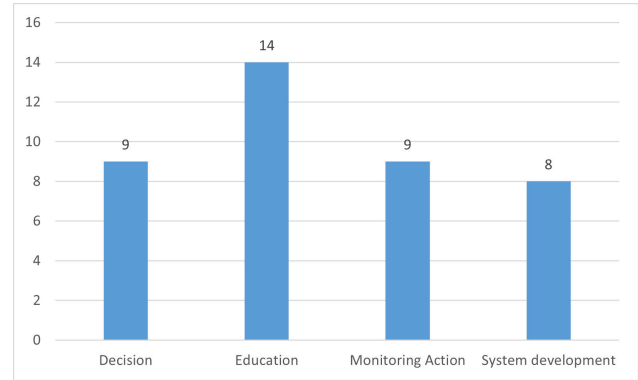


FIGURE 6. Percentage of XR technology used per application.

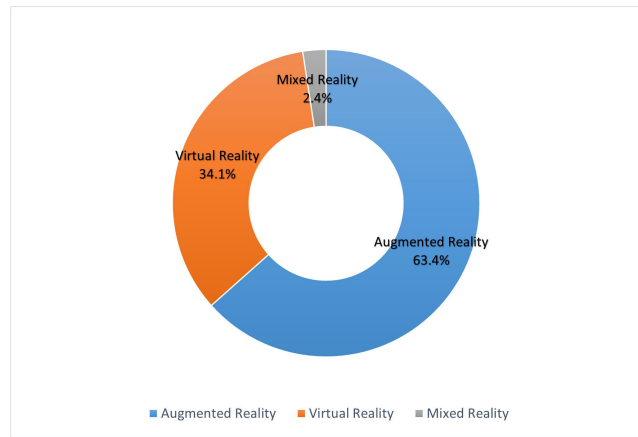


FIGURE 7. XR technology use per type (2022-2024).

divergence suggests that individual researchers are more actively pursuing AR applications. The consistently low representation of MR across both categories indicates its status as an emerging field with growth potential. These observations underscore a potential misalignment between the current research emphasis and the existing literature, which could influence future development priorities and the overall trajectory of immersive technology research. Figure 7 presents the findings, while Figure 8 displays the results from [22] findings.

2) XR TECHNOLOGY VIEWER TYPES

Head-mounted displays (HMDs) are now the predominant method of viewing content (59.1%). This represents a significant shift from earlier trends. Portable devices have become the second most common viewing method, used in 30.3% of the research studies, reflecting the increasing importance of mobile and handheld technologies in agricultural applications. Monitors, which were once the primary viewing method, are now used in only 10.6% of the studies. The significant reduction in monitor usage underscores the rapid adoption of more immersive and portable technologies in agricultural research and applications. The transition towards HMDs and portable devices indicates a strong trend towards more interactive, immersive, and field-applicable

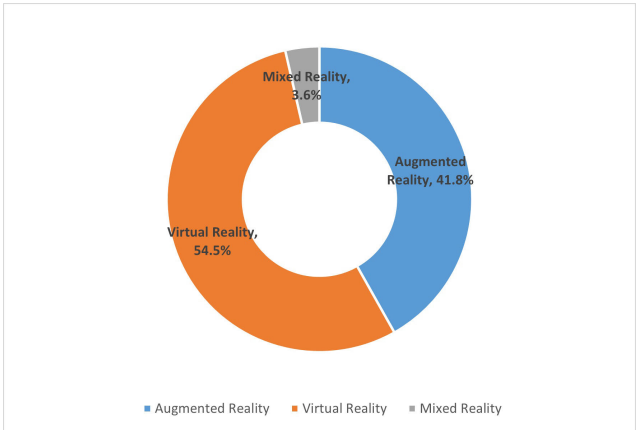


FIGURE 8. XR technology use per type (2001-2021) [22].

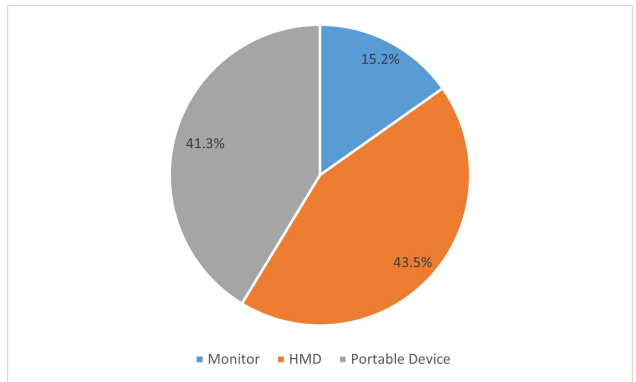


FIGURE 9. XR technology use per viewing device (2022-2024).

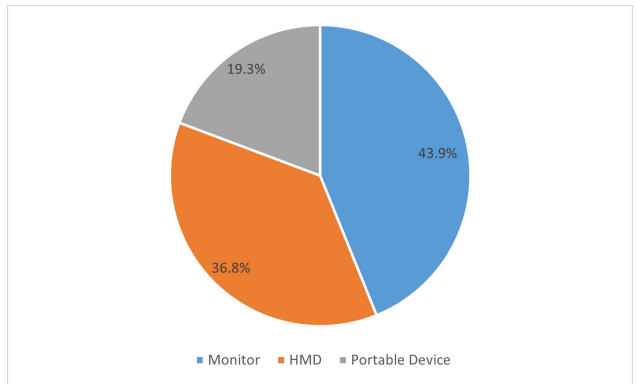


FIGURE 10. XR technology use per viewing device (2001-2021) [22].

technologies in agricultural studies and implementations. Figure 9 and Figure 10 illustrate the usage trends of these devices.

IV. DISCUSSION

A. XR RESEARCH IN AGRICULTURE AREA

Recent studies show a rapid increase in the use of XR technologies across agriculture, livestock farming, and aquaculture. This growth is evident in various exploration projects and proof-of-concept applications. For example, the “4D

Virtual Farm” platform highlights the rising interest in XR for agricultural education [29], while [30] created a virtual farm simulation to inform the public about the dairy industry. Although much of the current research consists of conference papers and early-stage projects, there are signs that the field is evolving. One notable advancement is the Vid2Cuts framework, which employs Artificial Intelligence (AI) to assist with grapevine pruning, signaling a move toward more sophisticated and practical XR applications [31]. Among the focus areas, agriculture leads in XR research, likely due to its larger scale and economic impact compared to livestock farming and aquaculture. The related applications span from crop management [32] to precision agriculture [38]. Education and training have emerged as key focus areas for XR in agriculture, mirroring trends in other fields. For instance, [74] and [75] explore the use of VR in agricultural education. Furthermore, XR also plays a role in decision-making support, as seen in [48]’s work on creating climate-smart and profitable farms using VR.

The development of more advanced systems, such as the real-time detection of strawberry ripeness using AR [43], reflects a growing trend toward combining XR with cutting-edge technologies like AI and the Internet of Things (IoT). In addition, cross-disciplinary applications are also on the rise, as illustrated by [35] use of AR and VR for immersive agricultural education through gamification.

While the field is advancing rapidly in agriculture areas, livestock farming areas and aquaculture areas have seen fewer studies. However, there is increasing interest in estrus detection and dairy cow identification using cascade deep learning for AR-based livestock management [56].

On the other hand, adapting AR and VR platforms for e-commerce to agriculture could revolutionize how farm products are marketed and sold. This feature allows consumers to virtually inspect produce and explore farms, potentially increasing farmers’ trust and sales. For instance, [89] applied augmented reality for on-site job training games to create immersive agricultural training simulations, enabling farm workers to safely practice operating machinery, manage crops, and handle livestock in virtual environments before applying these skills in real-world settings [90].

B. HUMAN-COMPUTER INTERACTION WITHIN AGRICULTURAL XR RESEARCH

Based on the findings, the trend in devices for viewing XR content has also shifted significantly toward HMDs and portable devices, with a noticeable decline in the use of traditional monitors. This shift is driven by the improved user experience provided by more immersive technologies. HMDs and portable devices, such as smartphones and tablets, allow for more natural and intuitive interaction with XR content. Reference [72] explores the potential of AR glasses in livestock farming, underscoring the benefits of wearable XR devices in real-world agricultural applications. The preference for HMDs and portable devices over

traditional monitors is partly due to the reduced occurrence of VR-induced symptoms like nausea and disorientation. These newer devices offer better motion tracking and lower latency, resulting in a more comfortable and prolonged user experience, an important factor in agricultural settings. Enhanced motion tracking and reduced latency can expand the applicability of XR technologies for tasks such as crop monitoring and livestock management. The advancement of ergonomic and user-friendly XR devices, exemplified by the work of [56] on AR applications in livestock farming, highlights the growing shift towards more immersive and portable XR solutions.

C. AGRICULTURAL XR TECHNOLOGIES AND APPLICATIONS

XR technologies have been applied across various agricultural applications, each catering to unique needs. Numerous studies, such as [1], [2], [5], [7], [10], [11], [12], [15], [19], [20], [29], and [32] have explored the integration of VR to create immersive simulations for education and farm planning. Noteworthy examples include the 4D Virtual Farm and the Grace platform, both designed to enrich agricultural education and enhance public awareness. Similarly, a virtual dairy farm simulation has been developed to introduce the public and high school students to agricultural practices. Beyond education, VR enables farmers to conceptualize climate-smart farms by allowing them to design and visualize operations within a fully controlled virtual environment.

AR has also emerged as a vital tool, particularly in crop management and precision agriculture, as illustrated in [3], [9], [13], [16], [17], [18], [21], [23], [24], [26], [27], [28], [29], [30], [31], and [91] studies. AR applications have been crafted to assist with real-time tasks, such as assessing strawberry ripeness, visualizing agrometeorological data in greenhouses, and monitoring plant growth. These innovations empower farmers to make informed decisions, optimize crop yields and enhance resource management. In addition, AR has been integrated with AI and the IoT to foster sophisticated, data-driven solutions, streamlining agricultural processes for heightened efficiency [3], [9], [14], [27], [33], [36].

Moreover, XR technologies have proven indispensable in livestock management, sustainable agricultural practices, and educational contexts. In livestock management, AR has been used for estrus detection and cow identification through advanced deep-learning techniques, while VR-based systems offer vocational training in fisheries and maritime industries. These applications support sustainable agriculture by helping farmers adapt to shifting environmental conditions and by promoting awareness of alternative food resources [11], [18], [20], [25]. In the realm of education, VR and AR technologies create immersive, interactive tools for both farm management and school curricula, reshaping the learning experience for students [1], [2], [7], [8], [16], [20], [22], [27], [28], [29], [30], [32]. These advancements highlight XR’s

transformative potential in agriculture, driving improved efficiency, sustainability, and the transfer of knowledge.

D. BENEFITS AND CHALLENGES OF XR ADOPTION IN AGRICULTURE

The integration of XR technologies in agriculture presents both significant opportunities and challenges. Research indicates that XR technologies offer substantial benefits in enhancing agricultural efficiency and productivity. According to [22]’s study, XR can provide real-time information, facilitate advanced training methodologies, and bridge knowledge gaps, particularly in remote agricultural regions. These technologies in precision agriculture contribute to sustainability by optimizing resource utilization and mitigating environmental impacts [76]. [78] emphasizes the potential of XR to create immersive experiences that revolutionize agricultural education, marketing strategies, and operational processes.

However, the adoption of XR in agriculture also faces some obstacles. Reference [81] identifies several challenges, including the complexity of 3D content creation, the substantial time investment required for staff training, and the need for robust implementation strategies to ensure long-term efficacy beyond the initial novelty. XR technologies also raise concerns regarding the potential mental, physical, and neurological effects of prolonged usage, particularly in educational contexts [74]. References [75] and [80] further discuss ongoing challenges in developing standardized interfaces, establishing guidelines for appropriate XR application across various agricultural scenarios, and seamlessly integrating XR technologies into existing educational and agricultural frameworks. These factors highlight the importance of a balanced approach to XR adoption, considering both its transformative potential and the practical and physiological factors involved.

E. ENVIRONMENTAL AND ECONOMIC IMPACT OF XR TECHNOLOGIES IN AGRICULTURE

XR technologies are poised to have significant environmental and economic impacts on the agricultural sector. These technologies enable farmers to make data-driven decisions, reducing the overuse of water, fertilizers, and pesticides, thereby lowering the ecological footprint of agricultural practices [80]. XR-based training and simulation tools can educate farmers on sustainable practices, fostering the adoption of environmentally friendly techniques [75].

One of the pressing issues in agriculture is the inefficient use of water, particularly in irrigated farming, which accounts for over 70% of global water use but operates with less than 45% efficiency. Reference [92] assessed the variability of Gross Biomass Water Productivity (GBWP) from 2009 to 2022. The integration of AR and VR technologies in precision agriculture can support the monitoring and optimization of Water Productivity (WP) by offering real-time data visualization, training farmers

in efficient irrigation practices, and facilitating simulations for improved water management strategies. Economically, these technologies in agriculture offer opportunities for increased productivity and cost reduction. For instance, [81] emphasizes the potential of virtual technologies to maintain quality in the agricultural sector. XR applications in farm management, as explored by [84], can streamline operations and reduce labour costs. Furthermore, XR’s use in agricultural education and training, as discussed by [74], can help cultivate a more skilled workforce, which could boost overall sector productivity. However, it is important to acknowledge that the initial investment in XR technologies and the necessary infrastructure development may pose economic challenges, particularly for small-scale farmers [70].

F. ETHICAL AND PRIVACY CONCERNS IN XR IMPLEMENTATIONS IN AGRICULTURE

The adoption of XR technologies in agriculture and farming causes four significant ethical and privacy challenges that require careful consideration. A primary concern involves the collection and management of sensitive farm data via XR devices. As AR and VR technologies are utilized to monitor crops, livestock, and equipment, they produce considerable amounts of proprietary data related to farming practices, yields, and land use [22], [93]. In the absence of robust security measures, this data is vulnerable to breaches or misuse, which could result in economic harm or competitive disadvantages for farmers.

There are also concerns regarding potential surveillance and breaches of worker privacy, as XR technologies can continuously track the activities and movements of farm labourers. This raises critical issues related to consent and data ownership [93], [94], [95]. Another ethical dilemma involves the risk of widening the digital divide, as smaller farms or those in developing regions may lack the resources necessary to adopt these technologies, placing them at a competitive disadvantage [96].

In addition, questions remain about the long-term effects of increased reliance on virtual environments for agricultural decision-making, which may distance farmers from their physical surroundings and traditional knowledge systems [22]. As XR technologies become more prevalent in agriculture, it is crucial to address these ethical and privacy concerns to ensure that the advantages of these innovations are realized without compromising the rights and well-being of farmers and workers.

G. CROSS-DISCIPLINARY COLLABORATIONS AND INNOVATION IN XR FOR AGRICULTURE

Cross-disciplinary collaborations in XR for agriculture are driving significant innovation and creating new opportunities for the sector. Reference [30] demonstrates this through their work on developing an immersive virtual farm simulation for public education about the dairy industry, which brings together experts from various fields. Similarly, [29] describes

the collaborative development of a Farm Systems Learning Platform called “4D Virtual Farm,” showcasing how diverse expertise can be combined to create innovative educational tools. Reference [32] discusses the development of an information system for modelling and visualizing crop growth under climate change conditions using 3D modelling and VR/AR technologies.

The integration of XR with other emerging technologies is opening up new frontiers in smart farming. Reference [38] proposes a seamless teleoperation pipeline for advanced VR-robot interfaces applied to precision agriculture. Reference [39] explores the use of VR-based digital twins in the context of pharmaceutical cannabis cultivation. These cross-disciplinary approaches, as evidenced by the work of researchers such as [44], are paving the way for more resilient, efficient, and sustainable agricultural practices. Reference [41] showcases the intersection of geographic information systems (GIS) and AR in their collaborative AR-GIS for land suitability visualization. The study brings together experts in geospatial technology, computer graphics, and agricultural planning to create a powerful tool for optimizing land use decisions.

In addition to the previously mentioned cross-disciplinary collaborations, [35] and [97] have made significant contributions to immersive agricultural education through their development of the GRACE platform. This innovative project combines expertise from agricultural science, game design, and educational technology to create an engaging learning experience. The platform utilizes both AR and VR technologies to gamify agricultural education, making complex concepts more accessible and interactive for students. By incorporating elements of game design into agricultural learning, they have demonstrated how interdisciplinary approaches can enhance educational outcomes and increase student engagement in agricultural topics.

H. EVALUATION OF XR TECHNOLOGY

Several studies have investigated the effectiveness and suitability of various systems, including those utilizing XR technology, emphasizing the importance of both qualitative and quantitative evaluations of the systems [98], [99]. These studies underscore the importance of conducting these evaluations to thoroughly assess the performance of AR and VR systems in educational settings utilizing surveys. However, an important avenue for future exploration lies in the deeper analysis of this qualitative feedback. Novel methodologies, such as computing with words and perceptual computing [100], [101], [102], offer promising tools for this purpose. These techniques allow for the computation of linguistic expressions from natural language, providing a more nuanced and contextual understanding of the feedback. Furthermore, some studies like Articulation [103] demonstrate how gamified tools can effectively support child-friendly assessments, offering valuable insights for similar applications in agriculture [104]. Integrating

these methodologies could lead to richer insights into the experiences and perspectives of farmers, students, teachers, and other stakeholders complementing traditional survey methods.

I. FUTURE DIRECTIONS AND RESEARCH GAPS IN XR FOR AGRICULTURE, LIVESTOCK FARMING, AND AQUACULTURE

The future of XR in the agriculture industry is filled with exciting growth opportunities, particularly in the realms of technological integration and educational applications. Research highlights a pressing need for improved interoperability between XR technologies and other emerging tools, such as AI and IoT. For instance, [31] proposes a framework for AI-guided vineyard management, which could be adapted for various agricultural tasks, illustrating how AI can enhance the precision and efficiency of farming practices. Furthermore, the development of immersive virtual farm simulations, as shown by [30], demonstrates XR’s potential to engage the public and educate them about sustainable practices, particularly in the dairy industry. By increasing the use of XR technologies in educational settings, we can significantly enhance the understanding of agricultural processes. Sustainability and climate change adaptation are also critical areas where XR can have a profound impact. Reference [32] highlights the value of 3D modelling and VR/AR technologies in visualizing crop growth under climate change scenarios, suggesting that future research should focus on leveraging XR for sustainable farm design and management. Reference [48] also illustrates how VR can play a key role in developing profitable and climate-smart farming strategies. These applications highlight the importance of incorporating XR into broader sustainability initiatives within the agricultural sector, addressing both environmental and economic challenges.

Despite these promising developments, there are still significant research gaps, particularly in the fields of livestock farming and aquaculture. While there have been advancements in AR for digital livestock management [46], further investigation is essential to fully grasp XR’s potential in promoting animal welfare and health monitoring. The aquaculture sector remains underrepresented in current XR research, creating an opportunity for future studies to delve into its applications in fish farming and aquatic ecosystem management. Therefore, it is vital to address user experience and adoption challenges, as noted by [40].

V. CONCLUSION

The integration of XR technologies in agriculture has demonstrated significant potential to revolutionize farming practices, education, and sustainability efforts. VR is creating immersive learning environments for agricultural education and farm planning, while AR is enhancing real-world farming activities through precision agriculture techniques. These technologies are being applied across various aspects of agriculture, from crop management and livestock monitoring to sustainable farming practices and public education. The

combination of XR with other advanced technologies like IoT, AI, and deep learning is further amplifying its impact, promising to improve farm productivity, enhance agricultural education, and contribute to more sustainable farming practices globally.

However, as XR technologies continue to evolve in the agricultural sector, several key concerns and areas for future development have emerged. These include 1) the need for seamless technological integration with other advanced systems, 2) emphasis on user-centred design to ensure accessibility for diverse user groups, 3) addressing data privacy and security concerns, 4) minimizing the environmental impact of XR devices and infrastructure, 5) improving scalability and cost-effectiveness for widespread adoption, 6) developing standardization and interoperability protocols, 7) conducting long-term impact assessments on agricultural practices and sustainability, and 8) addressing ethical considerations such as data ownership and fair access to technology. Addressing these concerns will be crucial for the successful and responsible implementation of XR technologies in agriculture, ensuring that their benefits are maximized while potential negative impacts are mitigated.

REFERENCES

- [1] D. B. Lobell, M. B. Burke, C. Tebaldi, M. D. Mastrandrea, W. P. Falcon, and R. L. Naylor, "Prioritizing climate change adaptation needs for food security in 2030," *Science*, vol. 319, no. 5863, pp. 607–610, Feb. 2008.
- [2] P. Gerland, A. E. Raftery, H. Ševšíková, N. Li, D. Gu, T. Spoorenberg, L. Alkema, B. K. Fosdick, J. Chunn, N. Lalic, G. Bay, T. Buettner, G. K. Heilig, and J. Wilmoth, "World population stabilization unlikely this century," *Science*, vol. 346, no. 6206, pp. 234–237, Oct. 2014.
- [3] M. M. Mekonnen and A. Y. Hoekstra, "Four billion people facing severe water scarcity," *Sci. Adv.*, vol. 2, no. 2, Feb. 2016, Art. no. e1500323.
- [4] M. González-Eguino, "Energy poverty: An overview," *Renew. Sustain. Energy Rev.*, vol. 47, pp. 377–385, Jul. 2015.
- [5] K. C. Seto, J. S. Golden, M. Alberti, and B. L. Turner, "Sustainability in an urbanizing planet," *Proc. Nat. Acad. Sci. USA*, vol. 114, no. 34, pp. 8935–8938, Aug. 2017.
- [6] S. Harper, "Economic and social implications of aging societies," *Science*, vol. 346, no. 6209, pp. 587–591, Oct. 2014.
- [7] M. Nicola, Z. Alsafi, C. Sohrabi, A. Kerwan, A. Al-Jabir, and C. Iosifidis, "The socio-economic implications of the coronavirus pandemic (COVID-19): A review," *Int. J. Surg.*, vol. 78, pp. 185–193, Oct. 2020.
- [8] C. K. Khoury, H. A. Achicanoy, A. D. Bjorkman, C. Navarro-Racines, L. Guarino, X. Flores-Palacios, J. M. M. Engels, J. H. Wiersma, H. Dempewolf, S. Sotelo, J. Ramírez-Villegas, N. P. Castañeda-Álvarez, C. Fowler, A. Jarvis, L. H. Rieseberg, and P. C. Struik, "Origins of food crops connect countries worldwide," *Proc. Roy. Soc. B, Biol. Sci.*, vol. 283, no. 1832, Jun. 2016, Art. no. 20160792.
- [9] P. Milgram and F. Kishino, "A taxonomy of mixed reality visual displays," *IEICE Trans. Inf. Syst.*, vol. 77, no. 12, pp. 1321–1329, Dec. 1994.
- [10] R. Azuma, "A survey of augmented reality," *Presence, Teleoperators Virtual Environ.*, vol. 6, no. 4, pp. 355–385, Aug. 1997.
- [11] P. Cipresso, I. A. C. Giglioli, M. A. Raya, and G. Riva, "The past, present, and future of virtual and augmented reality research: A network and cluster analysis of the literature," *Frontiers Psychol.*, vol. 9, p. 2086, Nov. 2018.
- [12] C. Flavián, S. Ibáñez-Sánchez, and C. Orús, "The impact of virtual, augmented and mixed reality technologies on the customer experience," *J. Bus. Res.*, vol. 100, pp. 547–560, Jul. 2019.
- [13] E. Z. Barsom, M. Graafland, and M. P. Schijven, "Systematic review on the effectiveness of augmented reality applications in medical training," *Surgical Endoscopy*, vol. 30, no. 10, pp. 4174–4183, Oct. 2016.
- [14] H.-K. Wu, S. W.-Y. Lee, H.-Y. Chang, and J.-C. Liang, "Current status, opportunities and challenges of augmented reality in education," *Comput. Edu.*, vol. 62, pp. 41–49, Mar. 2013.
- [15] T. Masood and J. Egger, "Augmented reality in support of Industry 4.0—Implementation challenges and success factors," *Robot. Comput. Integr. Manuf.*, vol. 58, pp. 181–195, Aug. 2019.
- [16] N. Gavish, T. Gutiérrez, S. Webel, J. Rodríguez, M. Peveri, U. Bockholt, and F. Tecchia, "Evaluating virtual reality and augmented reality training for industrial maintenance and assembly tasks," *Interact. Learn. Environ.*, vol. 23, no. 6, pp. 778–798, Nov. 2015.
- [17] D. C. Rose, R. Wheeler, M. Winter, M. Lobley, and C.-A. Chivers, "Agriculture 4.0: Making it work for people, production, and the planet," *Land Use Policy*, vol. 100, Jan. 2021, Art. no. 104933.
- [18] A. Y. C. Nee and S. K. Ong, "Virtual and augmented reality applications in manufacturing," *IFAC Proc. Volumes*, vol. 46, no. 9, pp. 15–26, Jan. 2013.
- [19] K. Liakos, P. Busato, D. Moshou, S. Pearson, and D. Bochtis, "Machine learning in agriculture: A review," *Sensors*, vol. 18, no. 8, p. 2674, 2018.
- [20] S. Wolfert, L. Ge, C. Verdouw, and M.-J. Bogaardt, "Big data in smart farming—A review," *Agricult. Syst.*, vol. 153, pp. 69–80, May 2017.
- [21] S. Doolani, C. Wessels, V. Kanal, C. Sevastopoulos, A. Jaiswal, H. Nambiappan, and F. Makedon, "A review of extended reality (XR) technologies for manufacturing training," *Technologies*, vol. 8, no. 4, p. 77, Dec. 2020.
- [22] E. Anastasiou, A. T. Balafoutis, and S. Fountas, "Applications of extended reality (XR) in agriculture, livestock farming, and aquaculture: A review," *Smart Agricult. Technol.*, vol. 3, Feb. 2023, Art. no. 100105.
- [23] D. Moher, A. Liberati, J. Tetzlaff, and D. G. Altman, "Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement," *PLoS Med.*, vol. 6, no. 7, Jul. 2009, Art. no. e1000097.
- [24] S. H.-W. Chuah, "Why and who will adopt extended reality technology? Literature review, synthesis, and future research agenda," *SSRN Electron. J.*, 2018.
- [25] M. Vasarainen, S. Paavola, and L. Vetoshkina, "A systematic literature review on extended reality: Virtual, augmented and mixed reality in working life," *Int. J. Virtual Reality*, vol. 21, no. 2, pp. 1–28, Oct. 2021.
- [26] S. Ke, F. Xiang, Z. Zhang, and Y. Zuo, "A enhanced interaction framework based on VR, AR and MR in digital twin," *Proc. CIRP*, vol. 83, pp. 753–758, Jan. 2019.
- [27] V. Pereira, T. Matos, R. Rodrigues, R. Nóbrega, and J. Jacob, "Extended reality framework for remote collaborative interactions in virtual environments," in *Proc. Int. Conf. Graph. Interact. (ICGI)*, Nov. 2019, pp. 17–24.
- [28] C. M. Andrews, M. K. Southworth, J. N. A. Silva, and J. R. Silva, "Extended reality in medical practice," *Current Treatment Options Cardiovascular Med.*, vol. 21, no. 4, Mar. 2019.
- [29] E. Hallein, D. Shallicross, J. Dalvean, P. Celi, M. McGowan, C. Jacobson, E. Bramley, J. F. Weston, and S. Barber, "Collaborative development of a farm systems learning platform '4D virtual farm,'" *J. Veterinary Med. Educ.*, vol. 52, no. 1, pp. 8–16, Feb. 2025.
- [30] A. Nguyen, M. Francis, E. Windfeld, G. Lhermie, and K. Kim, "Developing an immersive virtual farm simulation for engaging and effective public education about the dairy industry," *Comput. Graph.*, vol. 118, pp. 173–183, Feb. 2024.
- [31] S. Häring, S. Folawiyo, M. Podguzova, S. Krauß, and D. Stricker, "Vid2Cuts: A framework for enabling AI-guided grapevine pruning," *IEEE Access*, vol. 12, pp. 5814–5836, 2024.
- [32] O. A. Ivashchuk, V. A. Dunaeva, and I. I. Goncharova, "Development of an information system for modeling and visualization of crops growth and development under climate change conditions using 3D modeling and VR/AR technologies," in *Proc. Int. Russian Smart Ind. Conf. (SmartIndustryCon)*, Mar. 2024, pp. 161–166.
- [33] M. Fominykh and E. Prasolova-Forl, "Supporting career guidance and vocational training in fishery and maritime professions with virtual reality," in *Proc. Practitioner 10th Int. Conf. Immersive Learn. Res. Netw. (iLRN)*, 2024, pp. 69–71.
- [34] Y.-H. Hsieh and H.-C. Tsai, "Using virtual reality to enhance agricultural value of camellia formosensis: A native species in Taiwan," in *Proc. IEEE 7th Eurasian Conf. Educ. Innov. (ECEI)*, Jan. 2024, pp. 355–358.

- [35] M. Bigonah, F. Jamshidi, and D. Marghitu, "Immersive agricultural education: Gamifying learning with augmented reality and virtual reality," in *Cases on Collaborative Experiential Ecological Literacy for Education*, I. Global, Ed., Hershey, PA, USA: IGI Global, 2024, pp. 26–76.
- [36] Y. M. Rafee, D. Maying, and S. N. A. Leong, "The application of an augmented reality offline system for rural tourism development at Pa'Lungan, Bario," *Kajian Malaysia, J. Malaysian Stud.*, vol. 42, no. 2, pp. 1–32, 2023.
- [37] S. Raghav, K. Samridhi, and S. Khurana, "Applicability of augmented reality in agriculture using IoT application," *SSRN Electron. J.*, 2024.
- [38] M. Abdulsalam and N. Aouf, "VitRob pipeline: A seamless teleoperation pipeline for advanced virtual reality–robot interface applied for precision agriculture," in *Proc. IEEE Int. Conf. Robot. Biomimetics (ROBIO)*, Dec. 2023, pp. 1–6.
- [39] O. Spyrou, W. Hurst, and C. Verdouw, "Virtual reality-based digital twins: A case study on pharmaceutical cannabis," *Big Data Cognit. Comput.*, vol. 7, no. 2, p. 95, May 2023.
- [40] I. Asghar, R. Ullah, M. G. Griffiths, W. Warren, L. Dando, and J. Davies, "A user-centered system usability evaluation of a virtual reality application developed for solar farm training," in *Proc. 9th Int. Conf. Comput. Technol. Appl.*, May 2023, pp. 157–165.
- [41] H. Maulana and H. Kanai, "Designing collaborative augmented reality geographic information system for land suitability visualization," *J. Eng. Sci. Technol.*, vol. 18, pp. 2708–2721, Oct. 2023.
- [42] N. C. Weerasinghe, A. Avghs, W. Fernando, P. Rajapaksha, S. Siriwardana, and M. Nadeeshani, "HABARALA—A comprehensive solution for food security and sustainable agriculture through alternative food resources and technology," in *Proc. IEEE 17th Int. Conf. Ind. Inf. Syst. (ICIIS)*, Aug. 2023, pp. 175–180.
- [43] J. J. K. Chai, J.-L. Xu, and C. O'Sullivan, "Real-time detection of strawberry ripeness using augmented reality and deep learning," *Sensors*, vol. 23, no. 17, p. 7639, Sep. 2023.
- [44] K. Singh and M. Saini, "Towards digital twin of crops for growth modelling using virtual reality," in *Proc. ACM Multimedia Asia*, Dec. 2023, pp. 1–5.
- [45] M. Abinaya and G. Vadivu, "Cultivating a green future: Augmented reality veggie and fruit explorer app empowering Indian curriculum with agricultural knowledge and healthy habits," in *Proc. Int. Conf. Res. Methodolog. Knowl. Manage., Artif. Intell. Telecommun. Eng. (RMKMATE)*, Nov. 2023, pp. 1–8.
- [46] D. Pinna, G. Sara, G. Todde, A. S. Atzori, V. Artizzu, L. D. Spano, and M. Caria, "Advancements in combining electronic animal identification and augmented reality technologies in digital livestock farming," *Sci. Rep.*, vol. 13, no. 1, p. 18282, Oct. 2023.
- [47] N. A. Zahrein and N. Kamarudin, "Development of mobile durian farming application using augmented reality (AR)," *Appl. Inf. Technol. Comput. Sci.*, vol. 4, no. 2, pp. 532–551, 2023.
- [48] R. Lasseur, S. Laurenson, M. Ali, I. Loh, and M. Mackay, "Designing profitable and climate-smart farms using virtual reality," *PLoS ONE*, vol. 18, no. 6, Jun. 2023, Art. no. e0286723.
- [49] P. Piroozfar, A. Judd, S. Boseley, A. Viljoen, F. Castronovo, and E. R. Farr, "Enhancing user experience through VR: Case of an urban agriculture participatory design experiment," in *Proc. 13th Int. Conf. Construct. 21st Century*, Arnhem, The, Netherlands, 2023.
- [50] M. Ulagammai and R. N. Moorthy, "An augmented reality based intelligent precision agriculture using cascade advancement technique," in *Proc. 7th Int. Conf. Trends Electron. Informat. (ICOEI)*, Apr. 2023, pp. 303–307.
- [51] E. Romano, C. Bisaglia, A. Lazzari, A. Filisetti, E. Premoli, and M. Brambilla, "Augmented reality for the management of microclimate variability in greenhouses," in *Proc. IEEE Int. Workshop Metrology Agricult. Forestry (MetroAgriFor)*, Nov. 2023, pp. 811–814.
- [52] N. Capece, G. Manfredi, G. Gilio, U. Erra, F. Toscano, C. Fiorentino, and P. D'Antonio, "An educational approach for mixed reality visualization of agro-meteorological parameters," in *Proc. IEEE Int. Conf. Metrol. eXtended Reality, Artif. Intell. Neural Eng. (MetroXRINE)*, Oct. 2023, pp. 46–51.
- [53] S. Tamura, P. Buayai, and X. Mao, "BerryScope: AR strawberry picking aid," in *Proc. IEEE Int. Symp. Mixed Augmented Reality Adjunct (ISMAR-Adjunct)*, Oct. 2023, pp. 118–121.
- [54] C. Zeidler, M. Klug, G. Woelckner, U. Clausen, and J. Schöning, "ARCHIE2: An augmented reality interface with plant detection for future planetary surface greenhouses," in *Proc. IEEE Int. Symp. Mixed Augmented Reality (ISMAR)*, Oct. 2023, pp. 601–610.
- [55] H. Xing, F. Deng, Y. Tang, Q. Li, and J. Zhang, "The application of augmented reality technology in urban greening plant growth state detection," *IEEE Access*, vol. 11, pp. 59286–59296, 2023.
- [56] I. Arian, T. Ayav, A. Ç. Seçkin, and F. Soygazi, "Estrus detection and dairy cow identification with cascade deep learning for augmented reality-ready livestock farming," *Sensors*, vol. 23, no. 24, p. 9795, Dec. 2023.
- [57] T. Napier and I. Lee, "Using mobile-based augmented reality and object detection for real-time abalone growth monitoring," *Comput. Electron. Agricult.*, vol. 207, Apr. 2023, Art. no. 107744.
- [58] A. Soltiyeva, W. Oliveira, A. Madina, S. Adilkhan, M. Urmanov, and J. Hamari, "My Lovely Granny's Farm: An immersive virtual reality training system for children with autism spectrum disorder," *Educ. Inf. Technol.*, vol. 28, pp. 16887–168907, Dec. 2023.
- [59] I. K. A. G. Wiguna, R. Sanusi, I. G. I. Sudipa, K. N. Semadi, I. B. A. I. Iswara, I. P. A. E. D. Udayana, and M. D. W. Aristana, "Pengabdian berbasis teknologi dalam menunjang pengalaman dan interaksi pengunjung terhadap destinasi wisata melalui aplikasi AR gamification," *Faedah, Jurnal Hasil Kegiatan Pengabdian Masyarakat Indonesia*, vol. 1, no. 3, Aug. 2023.
- [60] D. V. Goncharov, O. A. Ivashchuk, and N. G. Reznikov, "Method for modeling and visualization of agricultural crops growth based on augmented reality technology in terms of the greenhouse effect dynamics," in *Proc. Int. Russian Smart Ind. Conf. (SmarIndustryCon)*, Mar. 2023, pp. 66–71.
- [61] L. L. Underwood, "Using virtual reality to increase student understanding and interest in farm animal welfare," *Tech. Rep.*, 2023.
- [62] A. Soni, A. Khare, P. S. D. Balaji, S. Verma, K. P. A. Rani, and S. Gowrishankar, "Pest identification and control using deep learning and augmented reality," in *Proc. Int. Conf. Edge Comput. Appl. (ICECA)*, Oct. 2022, pp. 1–7.
- [63] D. Neththikumarage, L. B. Liyanage, S. M. H. N. Samarakoon, P. M. Ramkumar, S. Harshanath, and U. U. S. Rajapaksha, "GARDENPRO—Android mobile application for home garden management using machine learning augmented reality and IoT," in *Proc. IEEE 10th Region Humanitarian Technol. Conf. (R10-HTC)*, Sep. 2022, pp. 300–305.
- [64] G. Styliaras, S. Tzima, and V. Dimou, "AR-based sustainability in food development," in *Proc. 13th Int. Conf. Inf., Intell., Syst. Appl. (IISA)*, Jul. 2022, pp. 1–8.
- [65] M. Zheng, N. V. Bermeo, and A. G. Campbell, "Augmented reality in-field observation creation and visualization in underperforming areas," in *Proc. IEEE Conf. Virtual Reality 3D User Interface Abstr. Workshops (VRW)*, Mar. 2022, pp. 852–853.
- [66] D. Roopa and S. Bose, "Leaf's diseases and its characteristics visualization using augmented reality," in *Proc. 1st Int. Conf. Comput. Sci. Technol. (ICCST)*, Nov. 2022, pp. 1019–1024.
- [67] E. Drutter, M. Milicevic, A. K. Skelin, and M. Bonkovic, "Augmented reality based sensor data visualization for plant growth monitoring," in *Proc. Int. Conf. Softw., Telecommun. Comput. Netw. (SoftCOM)*, Sep. 2022, pp. 1–7.
- [68] Y. Lai and M. Ismail, "Development of mobile application for applying aquaponic techniques in farming using augmented reality approach," *Appl. Inf. Technol. Comput. Sci.*, vol. 3, pp. 591–610, Nov. 2022.
- [69] K. Pavithra and A. Geetha, "Investigation of augmented reality in the agriculture industry and its future," in *Proc. Int. Conf. Recent Trends Comput.* Singapore: Springer, Jan. 2024, pp. 573–583.
- [70] M. Á. López-Cabarcos and J. Piñeiro-Chousa, "Illusion or reality? Building a metaverse community focused on value creation in the agricultural sector," *Int. J. Inf. Manage.*, vol. 77, Aug. 2024, Art. no. 102782.
- [71] A. Maitra and M. Damle, "Revolutionizing plant health management with technological digital transformation to enhance disease control & fortifying plant resilience," in *Proc. 3rd Int. Conf. Innov. Technol. (INOCON)*, Mar. 2024, pp. 1–8.

- [72] G. Sara, D. Pinna, G. Todde, and M. Caria, "Augmented reality glasses applied to livestock farming: Potentials and perspectives," *AgriEngineering*, vol. 6, no. 2, pp. 1859–1869, Jun. 2024.
- [73] F. Ecer, İ. Yaran Ögel, H. Dinger, and S. Yüksel, "Assessment of metaverse wearable technologies for smart livestock farming through a neuro quantum spherical fuzzy decision-making model," *Expert Syst. Appl.*, vol. 255, Dec. 2024, Art. no. 124722.
- [74] J. Pulley, D. Jepsen, A. Bowling, and T. Kitchel, "School-based agricultural education teachers' lived experience of integrating virtual reality into their classroom," *J. Agricult. Educ.*, vol. 65, no. 1, pp. 151–175, Mar. 2024.
- [75] J. A. Greig, B. Colclasure, S. Rampold, T. Ruth, and T. Granberry, "Enhancing agricultural education through virtual reality: Facilitation, application, reflection, and measurement in the classroom," *Advancements Agricult. Develop.*, vol. 5, no. 2, pp. 64–80, Jan. 2024.
- [76] P. Ashoka, N. K. Singh, N. H. Sunitha, D. R. K. Saikanth, O. Singh, G. Sreekumar, and B. V. Singh, "Enhancing agricultural production with digital technologies: A review," *Int. J. Environ. Climate Change*, vol. 13, no. 9, pp. 409–422, Jul. 2023.
- [77] A. Kolekar, S. Shalgar, and I. Malawade, "Beyond reality: A study of integrating digital twins," *J. Phys., Conf. Ser.*, vol. 2601, no. 1, Sep. 2023, Art. no. 012030.
- [78] F. Nakhle and A. L. Harfouche, "Extended reality gives digital agricultural biotechnology a new dimension," *Trends Biotechnol.*, vol. 41, no. 1, pp. 1–5, Jan. 2023.
- [79] F. Büyükkakin and Ö. B. Soylu, "Metaverse: Transformation and future of agriculture," in *Metaverse: Technologies, Opportunities and Threats*. Singapore: Springer, 2023, pp. 333–355.
- [80] F. Chen, C. Sun, B. Xing, N. Luo, and H. Liu, "Agricultural metaverse: Key technologies, application scenarios, challenges and prospects," *Smart Agricult.*, vol. 4, pp. 126–137, Dec. 2022.
- [81] S. Kumari, V. G. Venkatesh, and Y. Shi, "Virtual technologies adoption to sustain quality in agricultural sector: An emerging economy perspective," *J. Cleaner Prod.*, vol. 388, Feb. 2023, Art. no. 135988.
- [82] X. Wang and Y. Li, "The potential and opportunities of virtual reality technology in cultural restoration for agricultural revitalization," *Tech. Rep.*, 2023, no. 10020.
- [83] C. Khansulivong, S. Wicha, and P. Temdee, "Adaptive of new technology for agriculture online learning by metaverse: A case study in faculty of agriculture, national university of laos," in *Proc. Joint Int. Conf. Digit. Arts, Media Technol. With ECTI Northern Sect. Conf. Electr., Electron., Comput. Telecommun. Eng. (ECTI DAMT NCON)*, Jan. 2022, pp. 428–432.
- [84] T. Sneha, B. Nethravathi, N. H. Shahapure, S. Nagashree, and S. S. Shashidhara, "Future agriculture farm management using augmented reality: A study," in *Proc. 4th Int. Conf. Cognit. Comput. Inf. Process. (CCIP)*, Dec. 2022, pp. 1–4.
- [85] A. Nigam, P. Kabra, and P. Doke, "Augmented reality in agriculture," in *Proc. IEEE 7th Int. Conf. Wireless Mobile Comput., Netw. Commun. (WiMob)*, Oct. 2011, pp. 445–448.
- [86] B.-Y. Jiang and C.-M. Wang, "Combining augmented reality and multi-user remote collaboration to improve sustainable agriculture and economy," in *Proc. IET Int. Conf. Eng. Technol. Appl. (IET-ICETA)*, Oct. 2022, pp. 1–2.
- [87] A. Schütz, K. Kurz, and G. Busch, "Virtual farm tours—Virtual reality glasses and tablets are suitable tools to provide insights into pig husbandry," *PLoS ONE*, vol. 17, no. 1, Jan. 2022, Art. no. e0261248.
- [88] G. Sara, G. Todde, and M. Caria, "Assessment of video see-through smart glasses for augmented reality to support technicians during milking machine maintenance," *Sci. Rep.*, vol. 12, no. 1, p. 15729, Sep. 2022.
- [89] A. Asadi and R. Hemadi, "Design of an AR/VR platform for e-commerce," in *Proc. ICTD Nat. Conf.*, 2018, pp. 1–18.
- [90] A.-R. Asadi and R. Hemadi, "Augmented reality game creator for on-site job training," in *Proc. 2nd Nat. 1st Int. Digit. Games Res. Conf., Trends, Technol., Appl. (DGRC)*, Nov. 2018, pp. 128–133.
- [91] M. Speicher, B. D. Hall, and M. Nebeling, "What is mixed reality?" in *Proc. CHI Conf. Human Factors Comput. Syst.*, May 2019, pp. 1–15.
- [92] F. Shahryar, T. Rejoice, E.-A. Hesham, M. P. Sachithra, and T. Nikolay, "Monitoring agricultural water productivity and its potential climatic driving forces over the Nile delta between 2009–2022," in *Proc. AGU Fall Meeting Abstr.*, 2022.
- [93] D. Jones, S. Ghasemi, D. Gračanin, and M. Azab, "Privacy, safety, and security in extended reality: User experience challenges for neurodiverse users," in *Proc. Int. Conf. Hum.-Comput. Interact.* Cham, Switzerland: Springer, Jan. 2023, pp. 511–528.
- [94] I. Eweoya, S. R. Okuboyejo, O. A. Odetunmbi, and B. O. Odu-sote, "An empirical investigation of acceptance, adoption and the use of E-agriculture in Nigeria," *Heliyon*, vol. 7, no. 7, Jul. 2021, Art. no. e07588.
- [95] E. Mohammadrezaei, S. Ghasemi, P. Dongre, D. Gračanin, and H. Zhang, "Systematic review of extended reality for smart built environments lighting design simulations," *IEEE Access*, vol. 12, pp. 17058–17089, 2024.
- [96] F. da Silveira, F. H. Lermen, and F. G. Amaral, "An overview of agriculture 4.0 development: Systematic review of descriptions, technologies, barriers, advantages, and disadvantages," *Comput. Electron. Agricult.*, vol. 189, Oct. 2021, Art. no. 106405.
- [97] M. Bigonah, F. Jamshidi, A. Pant, and D. Marghitu, "Work in progress: Grace platform: Enhancing pedagogy with gamified AR and VR in agriculture education," in *Proc. ASEE Annu. Conf. Expo.*, 2024.
- [98] P.-L. Benedick, J. Robert, and Y. L. Traon, "A systematic approach for evaluating artificial intelligence models in industrial settings," *Sensors*, vol. 21, no. 18, p. 6195, Sep. 2021.
- [99] K. Baratimehr, M. R. Moosavi, and H. Tahayori, "Measures for evaluating IT2FSs constructed from data intervals," *Appl. Soft Comput.*, vol. 136, Mar. 2023, Art. no. 110084.
- [100] J. M. Mendel, "The perceptual computer: The past, up to the present, and into the future," *Informatik-Spektrum*, vol. 41, no. 1, pp. 15–26, Feb. 2018.
- [101] J. M. Mendel and D. Wu, *Perceptual Computing*. Hoboken, NJ, USA: Wiley, 2010.
- [102] K. B. Mehr, H. Tahayori, and M. R. Moosavi, "Information preserving approach for encoding words into interval type-2 fuzzy sets," *Granular Comput.*, vol. 9, no. 3, p. 55, Sep. 2024.
- [103] G. Alsebayel, M. Nasri, C. P. Myers, G. M. Troiano, E. Hatamimajourmer, S. Ostadabbas, K. Allison, and C. Hartevel, "ArticuMotion: Towards assessing motor speech disorders via gamification," in *Proc. 23rd Annu. ACM Interact. Design Children Conf.*, Jun. 2024, pp. 232–247.
- [104] T. Corwin, M. Kosa, M. Nasri, C. Holmgård, and C. Hartevel, "The teaching efficacy of the Protégé effect in gamified education," in *Proc. IEEE Conf. Games (CoG)*, Aug. 2023, pp. 1–8.



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